# AN AI POWERED SOFTWARE-BASED FIRE DETECTION SYSTEM FOR CCTV PLATFORMS.

## FiremeX - TEST CASES

AUTOMATION

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|---|---|
| **Document ID** | QA-AUTOMATION 05 |
| **Project** | FiremeX |
| **Tool** | Selenium |
| **Language** | Python |
| **Task** | UI Testing Admin Pages |
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| Test case ID | Test data | Expected result | Test Status |
|---|---|---|---|
| TC_DB_001 | Dashboard notification icon | Clicking the notification icon opens the Alerts page. | Passed |
| TC_DB_002 | Notification icon with unviewed alerts | The icon displays a visible, positive unviewed-alert count. | Failed |
| TC_DB_003 | Dashboard, Live Feed, Incidents, Alerts, Users, Settings | Each sidebar option is clickable and navigates to its corresponding page. | Passed |
| TC_DB_004 | Live Feed sidebar item | After navigating to Live Feed, its sidebar item is highlighted. | Passed |
| TC_DB_005 | Signed-in user profile | The user's name and role, such as "Administrator," are visible in the sidebar footer. | Passed |
| TC_DB_006 | User profile control | The profile control is enabled and clickable. | Passed |
| TC_DB_007 | Dashboard summary cards | Camera count 0, risk level Elevated, active alerts 3, and 2 unacknowledged are visible. | Passed |
| TC_DB_008 | Recent Incidents status badges: Fire detected, Smoke detected, Resolved, Camera offline | Status colors are red, yellow, green, and gray, respectively. | Passed |
| TC_DB_009 | Recent Incidents "View all" link | Clicking the link opens the Incidents page. | Passed |
| TC_DB_010 | Recent Incidents rows and timestamps | The table shows five incidents, ordered newest first. | Passed |
| TC_DB_011 | Recent Incidents table headers | The table shows Time, Camera, Zone, AI Conf., and Status columns. | Passed |
| TC_DB_012 | Camera preview detection status | A detection status, such as Fire or Smoke, is visible in the preview. | Passed |
| TC_DB_013 | Camera preview camera identifier and zone | The preview shows a camera number and zone. | Passed |
| TC_DB_014 | Detected cameras in the preview | Camera entries are visible and include camera identifiers and zone information. | Passed |
| TC_DB_015 | Dashboard sidebar item | The Dashboard sidebar item is highlighted while the Dashboard page is active. | Passed |
| TC_DB_016 | Sidebar "Sign out" button | Clicking Sign out ends the session and returns the user to the Login page. | Passed |
| TC_DB_017 | Header live clock and alert icon | The `LIVE HH:MM:SS` time is visible beside the alert icon. | Failed |

## Evidence

```
PS D:\QA-FiremeX\UIAutomation> python adminDashboard.py
test_TC_DB_001_notification_icon_is_clickable (__main__.FiremexAdminDashboardTests.test_TC_DB_001_notification_icon_is_clickable) ... ok
test_TC_DB_002_notification_icon_shows_unviewed_count (__main__.FiremexAdminDashboardTests.test_TC_DB_002_notification_icon_shows_unviewed_count) ...
ERROR
test_TC_DB_003_sidebar_options_are_clickable (__main__.FiremexAdminDashboardTests.test_TC_DB_003_sidebar_options_are_clickable) ... ok
test_TC_DB_004_active_sidebar_item_is_highlighted (__main__.FiremexAdminDashboardTests.test_TC_DB_004_active_sidebar_item_is_highlighted) ... ok
test_TC_DB_005_user_name_and_role_are_visible_in_sidebar_footer (__main__.FiremexAdminDashboardTests.test_TC_DB_005_user_name_and_role_are_visible_in_
sidebar_footer) ... ok
test_TC_DB_006_user_profile_icon_is_clickable (__main__.FiremexAdminDashboardTests.test_TC_DB_006_user_profile_icon_is_clickable) ... ok
test_TC_DB_007_dashboard_static_card_details_are_visible (__main__.FiremexAdminDashboardTests.test_TC_DB_007_dashboard_static_card_details_are_visible
) ... ok
test_TC_DB_008_recent_incident_statuses_use_expected_colors (__main__.FiremexAdminDashboardTests.test_TC_DB_008_recent_incident_statuses_use_expected_
colors) ... ok
test_TC_DB_009_recent_incidents_view_all_link_is_clickable (__main__.FiremexAdminDashboardTests.test_TC_DB_009_recent_incidents_view_all_link_is_click
able) ... ok
test_TC_DB_010_recent_incidents_shows_five_most_recent_rows (__main__.FiremexAdminDashboardTests.test_TC_DB_010_recent_incidents_shows_five_most_recen
t_rows) ... ok
test_TC_DB_011_recent_incidents_shows_required_columns (__main__.FiremexAdminDashboardTests.test_TC_DB_011_recent_incidents_shows_required_columns) ..
. ok
test_TC_DB_012_camera_preview_shows_detection_status (__main__.FiremexAdminDashboardTests.test_TC_DB_012_camera_preview_shows_detection_status) ... ok
test_TC_DB_013_camera_preview_shows_camera_number_and_zone (__main__.FiremexAdminDashboardTests.test_TC_DB_013_camera_preview_shows_camera_number_and_
zone) ... ok
test_TC_DB_014_all_detected_cameras_have_preview_entries (__main__.FiremexAdminDashboardTests.test_TC_DB_014_all_detected_cameras_have_preview_entries
) ... ok
test_TC_DB_015_dashboard_sidebar_item_is_highlighted (__main__.FiremexAdminDashboardTests.test_TC_DB_015_dashboard_sidebar_item_is_highlighted) ... ok
test_TC_DB_016_sign_out_button_is_clickable (__main__.FiremexAdminDashboardTests.test_TC_DB_016_sign_out_button_is_clickable) ... ok
test_TC_DB_017_live_time_is_visible_next_to_alert_icon (__main__.FiremexAdminDashboardTests.test_TC_DB_017_live_time_is_visible_next_to_alert_icon) ..
. ok
```

```
======================================================================
ERROR: test_TC_DB_002_notification_icon_shows_unviewed_count (__main__.FiremexAdminDashboardTests.test_TC_DB_002_notification_icon_shows_unviewed_count)
----------------------------------------------------------------------
Traceback (most recent call last):
  File "D:\QA-Firemex\UIAutomation\adminDashboard.py", line 131, in test_TC_DB_002_notification_icon_shows_unviewed_count
    count = button.find_element(
        By.XPATH,
        ".//*[self::span or self::div][normalize-space(.) "
        "and string(number(normalize-space(.))) != 'NaN']",
    )
  File "C:\Users\ASUS\AppData\Local\Programs\Python\Python314\Lib\site-packages\selenium\webdriver\remote\webelement.py", line 528, in find_element
    return self._execute(Command.FIND_CHILD_ELEMENT, {"using": by, "value": value})["value"]
           ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ASUS\AppData\Local\Programs\Python\Python314\Lib\site-packages\selenium\webdriver\remote\webelement.py", line 504, in _execute
    return self._parent.execute(command, params)
           ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^
  File "C:\Users\ASUS\AppData\Local\Programs\Python\Python314\Lib\site-packages\selenium\webdriver\remote\webdriver.py", line 506, in execute
    self.error_handler.check_response(response)
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^
  File "C:\Users\ASUS\AppData\Local\Programs\Python\Python314\Lib\site-packages\selenium\webdriver\remote\errorhandler.py", line 232, in check_response
    raise exception_class(message, screen, stacktrace)
selenium.common.exceptions.NoSuchElementException: Message: no such element: Unable to locate element: {"method":"xpath","selector":".//*[self::span or self::div][
normalize-space(.) and string(number(normalize-space(.))) != 'NaN']"}
```

| Test Case ID | Test data | Expected result | Test Status |
|---|---|---|---|
| TC_LF_003 | Click Add Camera on the Live Feed page | Opens /admin/livefeed/add-device; the camera source selector and Add Camera submit button are visible. | Failed |
| TC_LF_004 | Empty form, then display name Server Room L3, Home Assistant source, available camera entity, and zone Warehouse A | Add Camera is disabled while required fields are empty and becomes enabled after required fields are completed. | Failed |
| TC_LF_005 | Open the Add Camera form and click Cancel | Form closes and the app returns to /admin/livefeed. | Passed |

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|---|---|---|---|
| TC_LF_006 | At least one enrolled camera; click its Enable AI or Disable AI button | The button label toggles and the camera card displays the corresponding AI enabled/disabled state. | Failed |
| TC_LF_007 | Submit the Add Camera form with required details missing | Form displays validation feedback, such as a required-field, camera-name, or IP-address message. | Failed |
| TC_LF_008 | Leave the display name empty and submit | Browser validation rejects submission, identifies the name as missing, and keeps the user on the add-device page. | Passed |
| TC_LF_009 | Unique camera name; Home Assistant source; available entity set by FIREMEX_TEST_CAMERA_ENTITY;<br>zone Warehouse A | Camera registration succeeds, returns to Live Feed, and the new camera appears in a camera card. | Failed |
| TC_LF_010 | Camera card with a critical or alert status | Live Feed displays an alert-related state for the camera. Note: the script checks for alert-related text but does not verify red styling. | Failed |
| TC_LF_011 | Camera card; click its Edit control | Edit control is visible and clickable. | Failed |
| TC_LF_012 | Camera card; click its Delete control | Delete control is visible and clickable. | Failed |
| TC_LF_013 | A camera card on Live Feed | Card displays identifying or status information, such as a camera ID, camera label, normal status, zone, or live status. | Failed |
| TC_LF_014 | One or more registered cameras | Camera cards are rendered on the Live Feed page. | Failed |
| TC_LF_017 | Active camera stream | A camera preview element is present and visible. | Passed |
| TC_LF_018 | Live Feed page | Live Feed navigation item is visible in the sidebar. Note: the script checks visibility, not active-state styling. | Passed |

## Evidence




| Test Case ID | Test data | Expected result | Test Status |
|---|---|---|---|
| TC_UM_001 | Click Pending Requests tab | Tab is clickable and receives the selected-tab styling. | Passed |
| TC_UM_002 | Active Operators tab; one or more active operator records | Each operator card displays details and the labels System Role, Status, and Joined Date, plus the Revoke Access button. If there are no active operators, the page shows “No Active Operators.” | Passed |
| TC_UM_003 | Active operator; click Revoke Access, then dismiss the confirmation | Confirmation appears; dismissing it preserves the operator card. | Passed |
| C_UM_004 | Active Operators tab and its displayed operator cards | Active Operators count matches the number of operator cards. | Passed |
| TC_UM_005 | Click Pending Requests tab | Pending Requests tab is clickable and receives the selected-tab styling. | Passed |
| TC_UM_006 | At least one pending request | The request’s Approve Access button is enabled and clickable. | Passed |

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|---|---|---|---|
| TC_UM_007 | At least one pending request | The request's Deny Request button is enabled and clickable. | Passed |
| TC_UM_008 | Pending Requests tab; one or more pending requests | Each request card displays Requested Role, Requested Date, Approve Access, and Deny Request. If no requests exist, the page shows "No Pending Requests." | Passed |
| TC_UM_009 | Pending Requests tab and its displayed request cards | Pending Requests count matches the number of request cards. | Passed |
| TC_UM_010 | Select each tab: Active Operators and Pending Requests | The selected tab is visibly highlighted with the expected active styling. | Passed |

## Evidence

